Multiphase Flows – Laboratory Session 3: VOF Filling of a Tank (VOFFOT)

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Methodology

The test case consists of a cubic tank (1 m side) filled via a square section inlet pipe (0.2 m side). Water is injected at $V_{\text{inj}} = 2.2$ m/s into a quiescent air atmosphere at $p_{\text{atm}} = 101\,325$ Pa, with surface tension $\sigma = 0.072$ N/m and contact angle $\theta = 90^\circ$ [1]. Three hexahedral meshes are compared: coarse ($\Delta x = 0.05$ m), fine ($\Delta x = 0.025$ m), and fine with Adaptive Mesh Refinement (AMR, $\Delta x_{\min} = 0.0125$ m at the interface). The flow is assumed laminar to simplify wall treatment [1]; however, the jet Reynolds number $Re = V_{\text{inj}} D_h / \nu \approx 4.4 \times 10^4$ indicates transitional to turbulent conditions, representing a known physical limitation in the splashing dynamics.

The Volume of Fluid (VOF) method is a Eulerian Interface Capturing Method (ICM) that tracks the gas–liquid interface implicitly through the liquid volume fraction $\alpha_l \in [0, 1]$, governed by:

$$\frac{\partial \alpha_l}{\partial t} + \nabla \cdot (\alpha_l \, \mathbf{u}) = 0 \tag{1}$$

where $\mathbf{u}$ is the shared velocity field with mixture properties $\rho = \alpha_l \rho_l + (1 - \alpha_l)\rho_g$. Surface tension is included via the Continuum Surface Force (CSF) model [2].

Discretisation and Time-Stepping. The Modified HRIC scheme is applied to Eq. (1), operating in pure compressive form for $\text{Co} \leq \text{Co}_L = 1.0$ and fully blended with the diffusive Upwind scheme for $\text{Co} \geq \text{Co}_U = 1.2$ [1]. First order implicit time integration is used, enabling the explicit VOF multi step option (2–10 inner α_l steps per flow step, $\text{Co}_{\text{VOF}} = 0.4$) for the fine and adaptive meshes. An adaptive time stepping strategy with a free surface CFL limit of 0.2 modulates Δt automatically [1]. The bounds are given in Table 1.

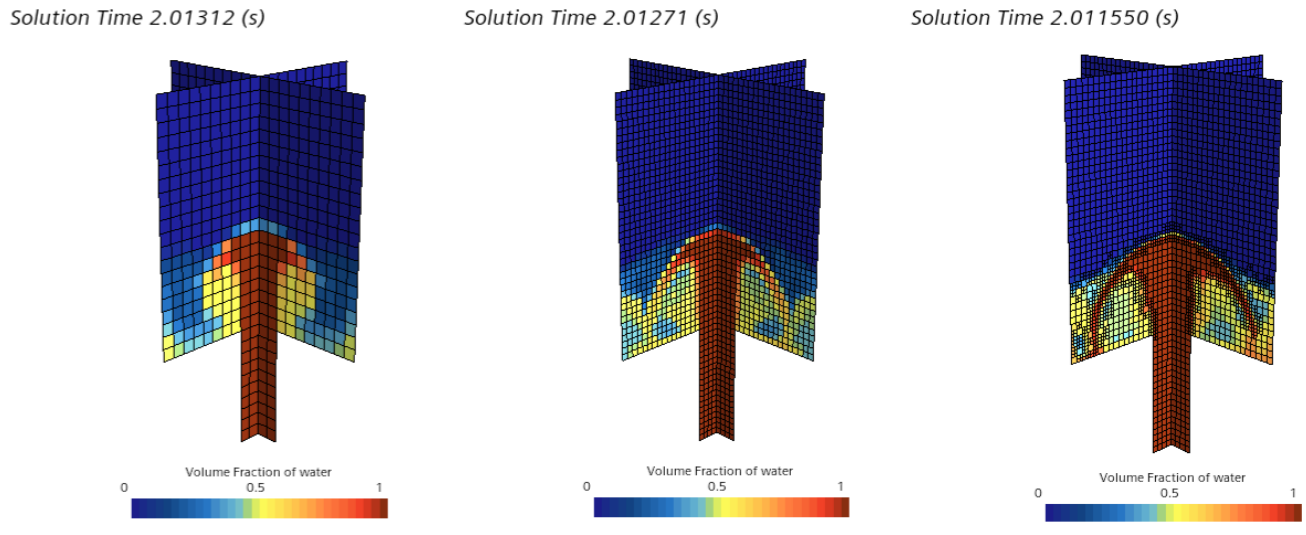
Boundary and Initial Conditions. The inlet is a velocity inlet ($V_{\text{inj}} = 2.2$ m/s, $\alpha_l = 1$); the top face a pressure outlet ($p = 0$ Pa gauge); walls are no slip with $\theta = 90^\circ$ [1]. Initially, liquid fills the lower half of the inlet pipe and the domain is at rest. The mass error rate model is enabled to monitor conservation.

Table 1: Time step bounds for $V_{\text{inj}} = 2.2$ m/s.

Mesh	$\Delta t_{\max}$ (s)	$\Delta t_{\min}$ (s)
Coarse	6.8×10^{-3}	6.8×10^{-4}
Fine	6.8×10^{-3}	6.8×10^{-4}
Adaptive	3.4×10^{-3}	3.4×10^{-4}

Results and Conclusions

Liquid Volume Fraction and Interface Smearing. Figure 1 compares mid-plane α_l contours at $t = 2$ s (the full temporal evolution across all three meshes is available in the companion video). The coarse mesh shows severe smearing: intermediate values of α_l spread over several cells, violating the sharp-interface requirement [2]. As a consequence, the surface tension force $\mathbf{F}_\sigma = \sigma \kappa \nabla \alpha_l$ loses physical meaning when $\nabla \alpha_l$ is artificially flattened, and the interface position becomes ambiguous. The fine mesh reduces this noticeably, while AMR achieves the sharpest interface by dynamically concentrating cells at the gas–liquid boundary, effectively halving the transition thickness relative to the fine mesh.



(a) Coarse **(b)** Fine **(c)** AMR

Figure 1: α_l mid-plane contours at $t = 2$ s. The jet core ($\alpha_l = 1$, red) and surrounding air ($\alpha_l = 0$, blue) are separated by an increasingly sharp interface with resolution; the coarse mesh smears this transition over $\sim$4 cells. Full evolution: video.

Figure 2 shows velocity vectors and streamlines in the mid plane at $t \approx 5.5$ s. The jet maintains a near uniform velocity profile through the pipe and decelerates sharply upon impacting the free surface, converting axial momentum into radial spreading. Air displaced by the rising liquid is recirculated downward along the tank walls, forming a large toroidal vortex. This two phase flow structure is only recoverable when the interface is sufficiently sharp to correctly couple the momentum of both phases. Peak velocities of 2.32 m/s at the pipe exit are consistent with V_{inj}, confirming correct boundary conditions.

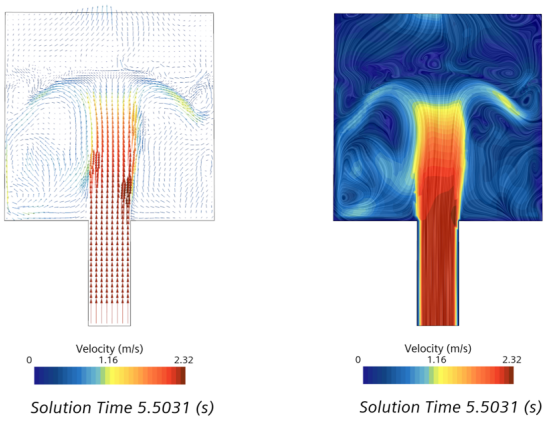
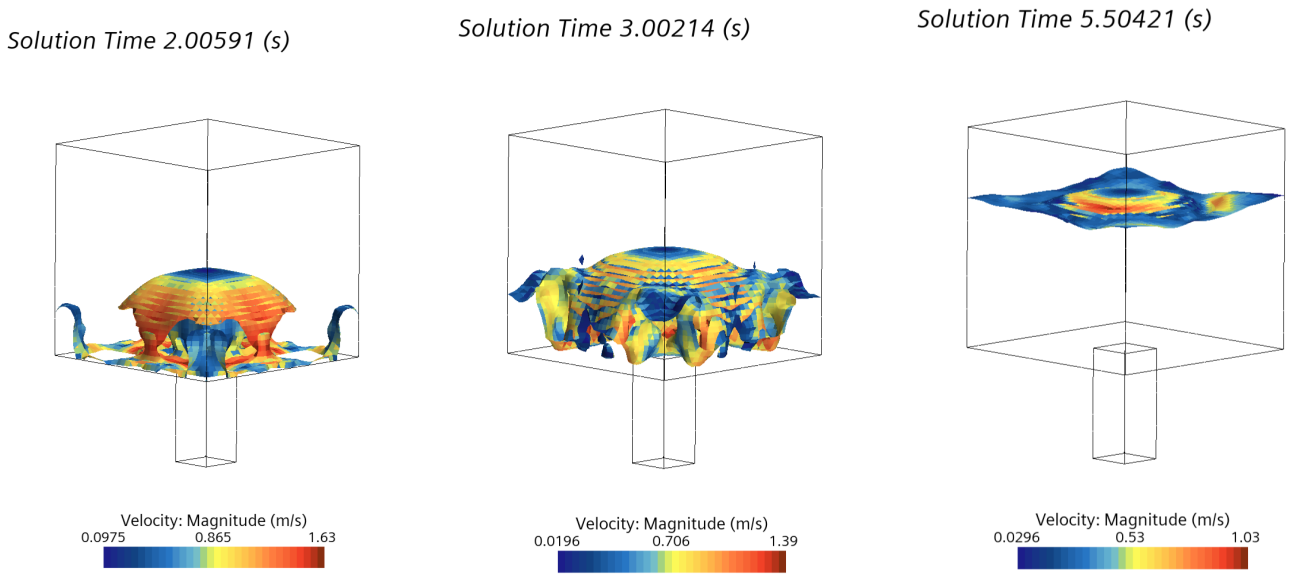


Figure 2: Velocity vectors (left) and streamlines (right) at $t = 5.503$ s, fine mesh. The jet decelerates at the free surface while air recirculates along the walls.

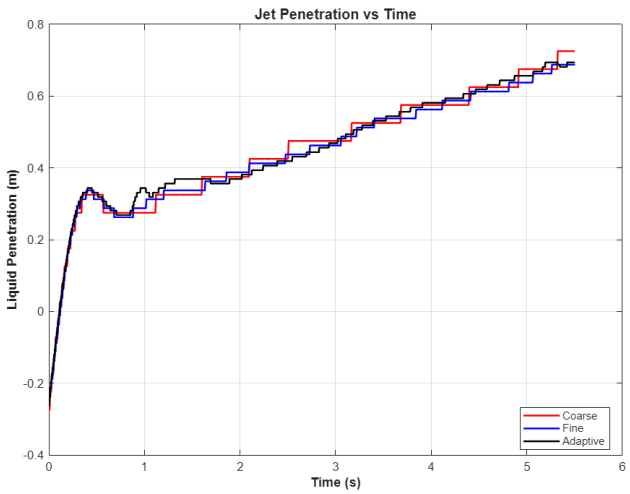
3D Interface Evolution (Fine Mesh). Figure 3 shows the $\alpha_l = 0.5$ iso-surface coloured by velocity magnitude at three solution times (continuous evolution available in the companion video). At $t \approx 2.0$ s a mushroom-shaped crown forms with peak velocities of 1.63 m/s at the apex; lateral lobes at the base suggest early Rayleigh–Taylor instability driven by the density contrast at the advancing front. The Courant number is below Co_L at this stage, so HRIC operates in compressive mode and the interface is well resolved. By $t \approx 3.0$ s the crown collapses under gravity and the liquid spreads radially; the wavy periphery reflects Kelvin–Helmholtz shear instabilities at the rim where liquid and air meet at high relative velocity. At $t \approx 5.5$ s the surface is nearly flat and mid-height in the tank, with dynamics driven by steady accumulation rather than jet momentum, and peak velocities drop to 1.03 m/s.



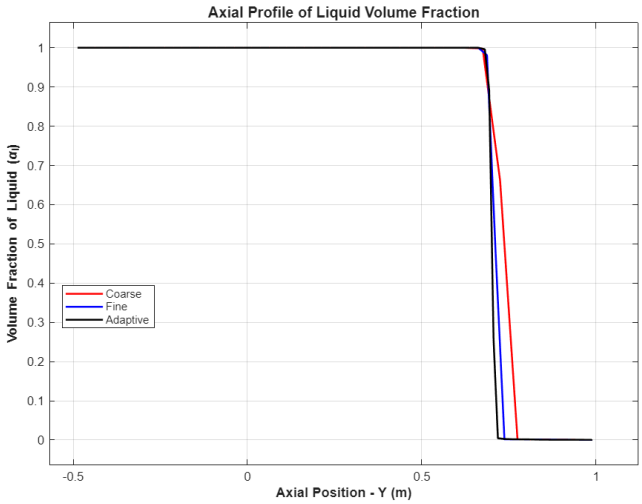
(a) $t = 2.006$ s **(b)** $t = 3.002$ s **(c)** $t = 5.504$ s

Figure 3: $\alpha_l = 0.5$ iso-surface coloured by velocity magnitude (m/s), fine mesh. (a) Crown with Rayleigh–Taylor lobes ($U_{\max} = 1.63$ m/s); (b) radial spreading with Kelvin–Helmholtz unstable rim ($U_{\max} = 1.39$ m/s); (c) quasi-flat surface at half-fill ($U_{\max} = 1.03$ m/s). Full evolution: video.

Quantitative Analysis: Penetration and Axial Profile. Figure 4 shows jet penetration over time and the axial α_l profile. The stepped shape of the penetration curves reflects the intermittent nature of jet's surface impact: the liquid front advances rapidly during each impact event and then pauses as the crown spreads laterally. During the initial phase ($t < 1.5$ s) all meshes agree well because the interface has not yet been deformed by impact. At later stages the coarse mesh progressively overestimates penetration (errors of up to 15% relative to AMR) because the diffuse α_l field artificially advances the $\alpha_l = 0.5$ threshold used as the penetration marker. The axial profile confirms this quantitatively: the liquid to gas transition spans approximately 3–4 cells in the coarse mesh but only 1–2 cells in the AMR mesh [2].



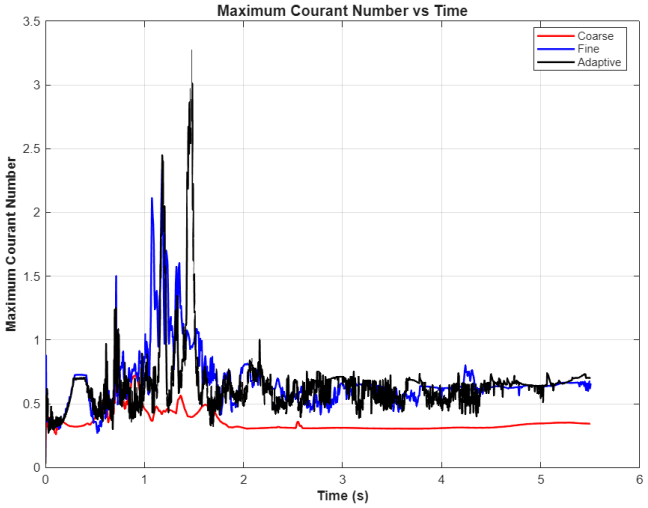
(a) Jet penetration vs. time



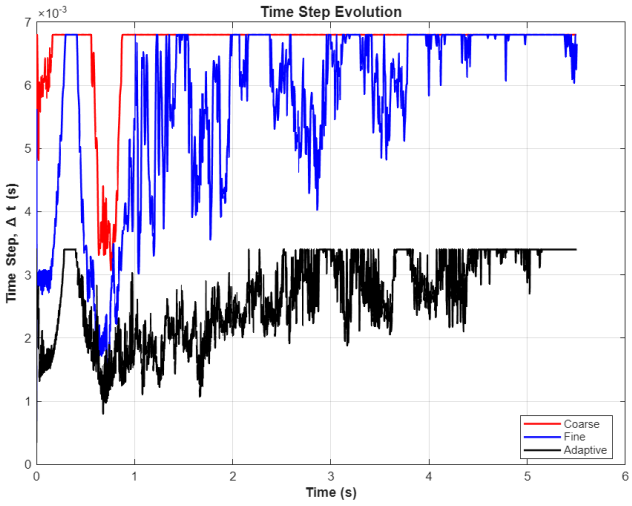
(b) Axial α_l profile

Figure 4: (a) Stepped curves reflect intermittent impact events; mesh divergence grows after $t = 1.5\,\mathrm{s}$ as smearing accumulates. (b) Axial profile: the coarse transition zone (3–4 cells) vs. the sharp AMR profile (1–2 cells) quantifies the smearing error.

VOF Validity: Courant Number and Adaptive Time-Stepping. Figure 5 reveals the key temporal limitation of the setup. The first Co spike occurs at $t \approx 1.8\,\mathrm{s}$, coinciding with the initial jet's surface impact when local velocities surge abruptly. The solver reduces Δt in response but rapidly hits $\Delta t_{\min}$ [1]; unable to reduce it further, Co spikes well above 3.5, forcing Star-CCM+ to deactivate HRIC and switch to the diffusive Upwind scheme. Subsequent spikes at $t \approx 2.5$ and $3.0\,\mathrm{s}$ correspond to secondary crown impacts. This is the root cause of the interface smearing observed even in the AMR mesh after each impact event. The time-step bounds in Table 1, scaled for steady pipe flow, were not conservative enough for the impulsive velocity peaks at impact; a $\Delta t_{\min}$ reduced by a factor of 5–10 would have kept Co below Co_U throughout.



(a) Maximum Courant number



(b) Adaptive time step (Δt)

Figure 5: Co spikes above 3.5 at $t \approx 1.8$, 2.5, and $3.0\,\mathrm{s}$ coincide with jet–surface impact events. Each spike forces the diffusive Upwind scheme as the solver hits $\Delta t_{\min}$ and cannot reduce the time step further.

Conclusions. The results confirm the sensitivity of the VOF method to both spatial and temporal resolution [2]. Three main conclusions emerge. First, AMR effectively sharpens the interface during stable filling phases, reducing the transition zone to 1–2 cells and providing the best accuracy at modest additional cost; it is therefore the recommended choice for engineering applications where interface fidelity is critical. Second, temporal resolution is equally important: the rigid $\Delta t_{\min}$ constraint causes Co exceedances during impact events in all meshes, forcing HRIC to revert to Upwind and producing unavoidable smearing that cannot be corrected by spatial refinement alone (a $\Delta t_{\min}$ reduced by a factor of 5–10 would eliminate this bottleneck). Third, the laminar flow assumption introduces a known physical inconsistency given $Re \approx 4.4 \times 10^4$; turbulence modelling [3] would improve the realism of the crown and splashing dynamics. Mass conservation was maintained in all cases, confirming numerical integrity regardless of interface sharpness.

References

[1] J. Nagy, *Multiphase (VOF) Simulation Project*, TU Wien, eulerian-solutions e.U., 2022.

[2] O. Ubbink, *Numerical prediction of two fluid systems with sharp interfaces*, Ph.D. thesis, Imperial College London, 1997.

[3] A. J. Torregrosa et al., "Study of turbulence in atomizing liquid jets," *Int. J. Multiphase Flow*, vol. 129, p. 103328, 2020.